

37. Case Study 2: The Hot Air Balloon (Applications of Trigonometry)

A Hot Air Balloon is flying at a height of 100 m above the ground. A man standing on the ground at point P observes the balloon at an angle of elevation of 60° . After some time, the balloon moves horizontally away from the point of observation, and the angle of elevation reduces to 30° .

Based on the above information, answer the following questions:

- (i) Draw a neat labelled diagram to represent the situation. (1 Mark)